

What Happens Next?

Video Action Prediction on Something-Something V2

Andrea Signoretti Florian Guillaumey

École Polytechnique — CSC_43M04_EP — 8 June 2026

andrea.signoretti@polytechnique.edu florian.guillaumey@polytechnique.edu

<https://github.com/FI0972/What-Happens-Next-Video-Action-Prediction-on-Something-Something-V2>

Abstract

We present in this report a two-track study of temporal action prediction on a 33-class subset of Something-Something V2 (SSv2) [4], “What Happens Next?”: only the first $\approx 40\%$ of each clip (four JPEG frames) is given, so the outcome is withheld and classification must reason about motion. In Track A (Closed World), where pretraining is forbidden, our study shows that where a model mixes time matters more than how much capacity it has: Temporal-Shift-Module [6] (TSM), which mixes frames before spatial pooling, beats VideoFormer-Lite (VFL, global post-pooling attention) by +3.0 pp at matched backbone size, and scaling its backbone from ResNet18 to ResNet50 then adds a further +3.4 pp to reach 42.85% val_dir (our best single model). A five-model log-softmax ensemble of TSM and VFL variants with test-time augmentation is our best submission and reaches 0.4785 public / 0.4547 private Kaggle top-1 score (16th). In Track B (Open World), we fine-tune V-JEPA 2 [1] and VideoMAE [8] backbones with progressive unfreezing, layer-wise learning-rate decay, window-capped source-frame re-extraction and pseudo-labelling, then build a 14-model uniform ensemble reaching 0.6586 Kaggle top-1 (16th also). We document a self-discovered data-leakage event (a discarded 0.81 submission) and show that ensemble diversity across architecture families beats single-family or learned-weight ensembling. Failure analysis attributes the residual errors to real-vs-pretended and direction-ambiguous classes that four frames under-determine.

1 Problem Description

Task. The challenge classifies four-frame JPEG clips into one of 33 motion-defined action classes (e.g. Pouring something into something, Pulling something from left to right). Crucially, each clip contains only the anticipatory portion of the source SSv2 video—the shipped frame_003 sits at a median fraction of 0.40 of the source clip—so the action’s outcome is deliberately withheld and

the task probes motion prediction, not final-state recognition [4].

Dataset. The split provides 44,993 training clips, 6,745 validation clips and 6,913 test clips over 33 classes (numeric prefixes 000–032; class 027 is absent from training). Classes are heavily imbalanced: the largest (009, Moving something up) has 3,170 examples while the smallest (026, Spilling something next to something) has only 162, a ratio of $19.6\times$. The KL divergence between training and validation class distributions is 0.088, confirming the splits are representative.

Evaluation and tracks. Submissions are scored by top-1 accuracy on a public+private Kaggle leaderboard over two stages (Stage 1, intermediate; Stage 2, final); top-5 is tracked internally. The provided baseline reaches ≈ 0.55 private and the second-place public reference is 0.74; our final submissions placed 16th in both tracks (counting the multiple accounts fielded by some teams). Following the challenge, we report two regimes. Track A (Closed World) prohibits pretrained models and external data—all spatial features, temporal dynamics and class boundaries are learned from scratch on the four provided frames. Track B (Open World) permits pretrained backbones and external corpora but not future frames or test labels. The task is hard for three reasons: (i) labels are motion-defined, defeating appearance-only models [4]; (ii) four frames is an order of magnitude below the 8–32 standard in video recognition [2, 6]; (iii) the discriminative cue—the action’s outcome—is withheld by construction.

2 Methods

2.1 Track A — Closed World

The pipeline evolved through several iterations on an internal training-split, all trained from scratch (the closed track forbids pretraining and external data). Our guiding hypothesis is that, without pretraining, every parameter spent on temporal modelling is one not spent learning spatial features, favouring parameter-free temporal mechanisms over heavy temporal heads — and that

where in the network temporal information is mixed matters as much as how it is mixed.

2.1.1 Architecture family and complexity

All three Track A architectures share the contract $(B, T, C, H, W) \rightarrow (B, 33)$ with $T=4$, $H=W=224$, and a ResNet backbone [5] mapping each frame to a 512-d (R18) or 2048-d (R50) vector after global average pooling (Fig. 1). CNNBaseline and CNN-LSTM simply average (resp. recurrently combine) the T pooled features — weak baselines (≈ 11.2 M params) since BPTT on four-step sequences barely helps. The two strong families differ in where they mix time relative to spatial pooling. TSM-ResNet [6] re-indexes, inside every residual block and before every convolution, a fraction $1/\text{fold_div}$ of channels one step backward and another $1/\text{fold_div}$ one step forward in time:

$$\text{TSM}(\mathbf{x})_{t,c} = \begin{cases} \mathbf{x}_{t-1,c} & c < C/4 \\ \mathbf{x}_{t+1,c} & C/4 \leq c < C/2 \\ \mathbf{x}_{t,c} & \text{otherwise,} \end{cases} \quad (1)$$

at zero parameter cost, while spatial resolution is still full. The T final per-frame vectors are averaged and a linear head $d \times 33$ yields the logits, so the total cost is exactly backbone + head: ≈ 11.2 M (R18) and ≈ 23.6 M (R50, 23.5 M of which is the backbone). TSM was designed for SSv2-style motion labels and is a strong prior here (Fig. 1).

VideoFormer-Lite (VFL) is our own design — a from-scratch-budget distillation of ViT [3] and TimeSformer [2] ideas that treats whole frames, not space-time patches, as tokens. A ResNet18 encodes each frame independently into a vector of size $d=512$, a learnable [CLS] token [3] is appended ($T+1=5$ tokens), and n_ℓ pre-norm Transformer blocks [9] (8 heads, ratio $r=4$) attend globally over them after pooling has already collapsed each frame to a single descriptor. Each block costs

$$P_{\text{block}} = \underbrace{4d^2}_{\text{Q,K,V,O}} + \underbrace{2rd^2}_{\text{MLP}} + \underbrace{4d}_{\text{LN}} \approx (4 + 2r)d^2 \approx 3.1\text{M}, \quad (2)$$

so $n_\ell=3$ adds ≈ 9.4 M to the ≈ 11.2 M backbone, for ≈ 20.7 M total — nearly twice TSM-ResNet18 for zero extra spatial capacity.

Where temporal mixing happens — the key difference. TSM mixes time at full spatial resolution, before pooling: by block 8, each frame’s representation already encodes a receptive field over all T frames at the (h, w) level. VFL pools first and attends only over $T+1$ frame-averaged descriptors — it never sees motion at the pixel level. On SSv2, where the cue is typically a single finger or edge moving within a small region, this matters: TSM keeps the spatial precision the cue lives in; VFL has already discarded it (Fig. 1). Sec. 3.1 measures this argument directly.

Technique	Setting	Source
Reg. warm-up	10 ep, no MixUp/CutMix	this work
MixUp / CutMix	$\alpha=0.2$ / $p=0.25$, after ep 10	[10, 11]
Focal loss	$\gamma \in \{0, 1, 2\}$ (best ≈ 1)	[7]
Label smoothing	$\varepsilon=0.1$	standard
RandAugment	2 ops, mag. 0.5	standard
Random erasing	$p=0.25$	standard
Horizontal flip	disabled	label semantics
Stochastic depth	DropPath 0.1	standard
Optimiser	AdamW, lr 10^{-3} , cosine	standard

Table 1. Track A training recipe (best single model: TSM-ResNet50, $T=4$, fold_div=8, rotating folds, $\sim 42.85\%$ val_dir).

2.1.2 Critical frame-count fix

The single largest gain (+4.4 pp) came from matching num_frames to the true clip length (four real frames). num_frames>4 forces linspace interpolation that duplicates frames, so consecutive identical tokens make Eq. (1) read the same value twice — zero temporal gradient on the shifted channels, corrupting the signal TSM relies on. Setting $T=4$ lifted tsm_ultra_v2 to 57.73% internal / 38.25% val_dir (+4.41/+3.34 pp); a first ResNet50 attempt at the old $T=8$ regressed to 45.26% internal (Fig. 3, Fig. 2) — capacity cannot undo corrupted input. Once $T=4$ was enforced, a rotating-fold ResNet50 run reached 42.85% val_dir, our best single model: fix the pipeline before scaling.

2.1.3 Regularisation recipe

All Track A models share a regularisation stack (Table 1) whose critical element is a warm-up delay: MixUp [11] and CutMix [10] are disabled for the first 10 epochs so the backbone forms basic per-frame representations before label-mixing starts — worth +36.3 pp on VFL alone ($9.45 \rightarrow 45.78\%$), since a randomly-initialised head cannot meaningfully mix labels it cannot yet distinguish, and early MixUp collapses the representation rather than regularising it. Two further choices are domain-specific: horizontal flip is disabled (it inverts SSv2’s direction-encoded labels, e.g. Pulling left $\rightarrow$ right), and CutMix probability is kept low ($p=0.25$, a cut patch spans all four frames and often occludes the defining motion).

2.1.4 Rotating k -fold and ensemble

The strongest checkpoints also have a rotating-fold variant retrained on every fold of the train+val union (held-out fold = epoch mod k , $k=5$): higher-bias snapshots contributing uncorrelated errors at inference (Fig. 10), worth +1.2 pp for TSM-ResNet18 and +4.3 pp for VFL (its data-hungry attention head lacks TSM’s structural temporal prior and so benefits more from extra data; Sec. 3.1 shows it saturates at $k=5$).

The final submission fuses five checkpoints by log-softmax late fusion — averaging in log-probability space

down-weights over-confident errors and rewards agreement among differently-calibrated members: three with 10-crop TTA (tsm_ultra_v2, its rotating-fold variant, and video_former_lite_ultra_rot), and two without (TSM-ResNet50 rotating, our best single model at 42.85%, and a CE-trained $\gamma=0$ TSM variant whose different loss geometry adds complementary calibration). It reaches 46.75% val_dir (Fig. 6) and 0.4785 public / 0.4547 private on Kaggle — our best closed-track score, +6.2 pp over the strongest single model (0.4163; Table 2). TTA alone adds ~ 1 pp on the strongest single model (Fig. 8), trading off on direction-encoded classes (Fig. 9) where hflip views push confidences toward the mirror class even after label-aware remapping.

2.2 Track B — Open World

Top SSv2 teams use video transformers pretrained on large unlabelled corpora. We selected VideoMAE [8] (masked-autoencoder pre-training; HuggingFace Kinetics-400 checkpoints) and V-JEPA 2 [1], a joint-embedding predictive architecture that reconstructs masked regions in feature space, focusing capacity on semantically relevant motion. We use the self-supervised vjepa2-vitl-fpc64-256 checkpoint, pretrained on unlabelled video without SSv2 labels—required for integrity (Sec. 3.4).

Progressive unfreezing with LLRD. V-JEPA is fine-tuned in three phases with layer-wise LR decay (block ℓ gets $\eta \cdot \lambda^\ell$, $\lambda=0.75$): a frozen attentive probe reaches 0.672 EMA internal val, unfreezing the top-6 blocks reaches 0.705, and full-encoder fine-tuning reaches 0.728 internal / 0.6411 val_dir. Freezing first stabilises the head and preserves pretrained feature geometry; the resulting staircase is visible in Fig. 12.

Window-capped source-frame re-extraction. V-JEPA benefits from 16 frames but only four are shipped; since folder names encode SSv2 video IDs, the source WebM files are reachable. We match the shipped frame_003 to source frames by pixel-level MAE to locate the provided window’s end, sample 16 frames within it, and hard-bound the search to the clip’s first 50% so the withheld outcome ($> 60\%$) stays unreachable by construction (median window fraction 0.39, max 0.50; Fig. 13).

Pseudo-labelling. Following noisy-student self-training, the Phase-2 ensemble soft-labelled the test clips; a ≥ 0.85 confidence threshold retained all 6,913 clips (the model is over-confident), so the loop effectively trained on $\text{train} \cup \text{test}$ under its own predictions. The resulting V-JEPA-pseudo reached 0.6519 val_dir / 0.6410 Kaggle, but its widened gap (Sec. 3) signalled over-fitting, so we stopped there.

Honest cross-preprocessing ensemble. The final system aggregates 14 checkpoints from three families, each on its own preprocessing (V-JEPA: 6, 16-frame 256 px window-capped; VideoMAE-Base/Large K400: 3+3, 4-frame

Trk	Method	val_dir	Kgl pub.	Kgl priv.
A	TSM-R18 (single, +TTA)	41.68%	0.4163	0.3881
A	TSM-R18 (single)	38.25%	0.3576	0.3442
A	VFL-Ultra (single)	32.11%	—	—
A	TSM-R50 rot. (single)	42.85%	—	—
A	3-model ens. +TTA	44.03%	0.4508	0.4260
A	5-model ens. (R50)	46.75%	0.4785	0.4547
B	V-JEPA-ft standalone	64.11%		0.6361
B	V-JEPA-pseudo	65.19%		0.6410
B	14-uniform	65.19%		0.6586

Table 2. Summary of Track A and Track B results (val_dir = internal top-1; Kgl = Kaggle public/private top-1, reported separately for Track A where the gap is informative, combined for Track B). Track A ResNet50 has no solo Kaggle submission — evaluated as an ensemble member only.

224px; TSM-ResNet50: 2, 4-frame), combined by uniform softmax averaging: diverse, distinct models produce uncorrelated errors that cancel, so even the weakest member (TSM-ResNet50) raises the ensemble (Sec. 3).

2.3 Reused vs. own code

Reused: the course starter repo (Hydra, VideoFrameDataset, train.py); torchvision ResNet [5] and R(2+1)D backbones; pretrained HuggingFace V-JEPA 2 [1] and VideoMAE [8] checkpoints (Track B only). Our own: the parameter-free TSM wrapper [6]; VideoFormer-Lite (Sec. 2.1.1); the regularisation-warm-up recipe, rotating k -fold protocol, ablation suite, and log-softmax ensemble/TTA pipeline; and the full Track B stack — window-capped re-extraction, progressive-unfreezing+LLRD, and noisy-student pseudo-labelling.

3 Results

Overall. Table 2 reports headline results with Kaggle scores for both tracks. Track A reaches 46.75% val_dir / 0.4785 Kaggle public from scratch; Track B reaches 0.6586 Kaggle top-1 (65.19% internal val) with the 14-model uniform ensemble — above the 0.55 baseline and close to the 0.74 second-place reference. The ≈ 18 -pp Kaggle gap between the tracks’ best submissions closely tracks the ~ 18 -pp val_dir gap: forgoing large-scale self-supervised pretraining costs about the same whether measured internally or on the independent Kaggle distribution — both tracks generalise with a comparable, small, positive offset (calibration below).

3.1 Track A ablations

Regularisation warm-up (+36.3 pp). Single variable: without warm-up VFL stalls at 9.45% (killed at epoch 4) and R(2+1)D at 10.78%; with a 10-epoch warm-up VFL reaches 45.78%. The effect transfers across architectures: a general property of cold-start training with label-mixing, not an architecture-specific quirk.

Frame count (+4.4 pp). $T=5 \rightarrow 4$ improves internal val by +4.41 pp (53.32 $\rightarrow$ 57.73) and val_dir by +3.34 pp (34.91 $\rightarrow$ 38.25); the first ResNet50 attempt at the old $T=8$ regressed ≈ 12 pp below the $T=4$ ResNet18 internal score (Fig. 3).

Architecture: TSM vs. VFL (+3.0 pp for local mixing). At matched settings (rotating folds, $T=4$, ResNet18, no TTA), TSM reaches 39.45% val_dir vs. VFL’s 36.44% (+3.01 pp). With all else fixed, the only difference is where the two mix time — before pooling (TSM) or after (VFL) — so this is the controlled test of Sec. 2.1.1’s argument, and local mixing wins clearly, as expected for labels defined by small, spatially-localised motion cues that pooling destroys.

Backbone scale-up, done right (+3.4 pp). With $T=4$ and rotating folds fixed, swapping ResNet18 for ResNet50 (11.2 $\rightarrow$ 23.6 M params) brings 39.45% $\rightarrow$ 42.85% val_dir (+3.40 pp, Fig. 2). Together with the $T=8$ regression above, this closes the loop on our central claim: capacity is useless on top of a corrupted temporal signal, but pays off once the signal is correct — the same $2.1\times$ parameter increase that wasted itself on duplicated frames at $T=8$ converts cleanly into accuracy at $T=4$.

Focal-loss γ sweep and VFL ablations (small, informative effects). With TSM-ResNet18 otherwise frozen, CE ($\gamma=0$) reaches 38.92%, $\gamma=1$ reaches 39.50%, $\gamma=2$ reaches 39.45% (Fig. 4): any focal weighting beats CE by ≈ 0.5 pp, but $\gamma \in [1, 2]$ is within noise — the gain comes from having a confusable-pair-aware loss, not tuning it finely. The CE model still adds +0.52 pp to the final ensemble (Sec. 2.1.4): a different loss geometry gives a different calibration, exactly what log-softmax fusion exploits. On VFL (Fig. 5), dropping from 3 to 2 Transformer layers (Eq. (2): -3.1 M) costs -1.08 pp (36.44 $\rightarrow$ 35.36%), and over-rotating from 5 to 10 folds costs a further -0.46 pp — VFL saturates its data-utilisation capacity at $k=5$, so extra coverage just adds gradient noise from smaller validation folds.

Ensemble components and leave-one-out. Table 3 (Fig. 7) leaves out one member at a time from the final 5-model ensemble (46.75%). Every member is necessary, but the margins are revealing: dropping the ResNet50 model costs -1.33 pp, far the largest drop — more than any same-backbone TSM-ResNet18 variant (-0.31 to -0.47 pp) or the architecturally distinct VFL (-0.64 pp). Backbone diversity contributes more than architecture-family diversity at matched capacity: TSM and VFL stay complementary, but the largest source of useful, decorrelated error is a model that simply sees more (deeper backbone), not one that sees differently.

3.2 Track B ablations

Honest Kaggle progression. Table 4 (Fig. 11) traces the key honest milestones. The 21 pp gap between the discarded leaky run (0.81) and the first honest baseline

Configuration	Top-1	Δ
Full ensemble (5 models)	46.75%	—
– TSM-ResNet50 (rot)	45.41%	-1.33
– VFL-Ultra-rot + TTA	46.11%	-0.64
– TSM-no-focal (rot)	46.23%	-0.52
– TSM-Ultra-v2-rot + TTA	46.27%	-0.47
– TSM-Ultra-v2 + TTA	46.43%	-0.31

Table 3. Track A leave-one-out on val_dir for the final 5-model ensemble (46.75%). Every component is necessary; the ResNet50 backbone is by far the dominant contributor, ahead of the architecturally distinct VFL.

Submission	Kaggle	Δ
Leaky V-JEPA (discarded)	0.81	—
V-JEPA-ft standalone (clean)	0.6361	—
9-uniform (+ K400 + TSM)	0.6418	+0.6
12-uniform (+ pseudo)	0.6546	+1.9
14-uniform (+ Large)	0.6586	+2.3

Table 4. Track B Kaggle submission history (Δ in pp vs. the first honest baseline). Diversity, not single-family scaling, drives every gain.

Strategy	n	Kaggle
V-JEPA-only, uniform	6	0.6410
9-uniform (+ K400 + TSM)	9	0.6418
V-JEPA-heavy (3:1 weight)	12	0.6499
12-uniform	12	0.6546
14-uniform	14	0.6586

Table 5. Track B ensemble-strategy ablation. Uniform averaging of diverse families beats single-family and learned/manual weighting.

(0.6361) quantifies what data leakage was buying; every later point came from legitimate architectural diversity. Fig. 14 plots Kaggle vs. ensemble size: returns stay positive at $n=14$, and the 6-model V-JEPA-only mix underperforms the 9-model uniform mix despite higher individual accuracies — a first hint that family diversity matters more than per-model strength.

Diversity beats weighting. Table 5 isolates strategy on a fixed pool: V-JEPA-only uniform (0.6410) underperforms the 12-model mix (0.6546) despite higher individual accuracy; manual 3:1 weighting (0.6499) lands in between. Uncorrelated errors drive the gain: TSM-ResNet50 (0.3364 val_dir) has the lowest agreement with every transformer family (0.37–0.38), while two V-JEPA variants agree at 0.85 — exactly why the weak member still helps. Learned weighting (Fig. 15) collapses 82% of mass onto two checkpoints and discards that decorrelation, the same lesson as Track A’s leave-one-out: the most valuable member is the most different.

Calibration. The val_dir–Kaggle gap is +0.005 for V-JEPA-ft but doubles to +0.011 for V-JEPA-pseudo

— the signal that stopped pseudo-label iterations. Track A’s best ensemble shows the opposite sign (46.75% vs. 0.4785, gap -0.011): from-scratch heavy-augmentation models generalise to the unseen test split at least as well as to their own validation, whereas fine-tuned models can over-specialise, widening the gap with each pseudo-label round.

3.3 Failure analysis

Both tracks share the same dominant error patterns, mirroring SSv2’s design intent (Figs. 16, 17, 18). (i) Real vs. pretended: the worst confusion is Picking something up $\rightarrow$ Pretending to pick something up (56/199 clips, $F1=0.07$): the early trajectory is identical and the disambiguating cue—whether the object actually rises—falls in the withheld future. (ii) Temporal direction: Folding $\leftrightarrow$ Unfolding (24–29 mutual errors) share appearance and differ only in motion direction, fragile at four frames. (iii) Static sink: Holding something absorbs misclassified motion clips (Moving down: 30; Opening: 28) as a default. Best classes ($F1 > 0.54$) are clean single-arc motions (Pulling left $\rightarrow$ right, Pouring, Uncovering, Folding). Class-size vs. accuracy correlation is weak ($r=0.24$): difficulty is semantic, not scarcity-driven.

3.4 Data-integrity

An early extraction sampled frames over the full clip (0–100%); training V-JEPA yielded 0.81 Kaggle, ~ 7 pp above second place (Fig. 11). Two leakage vectors: (1) frames from the withheld outcome ($> 60\%$) were reachable; (2) an initial V-JEPA checkpoint had been fine-tuned on SSv2 with ground-truth labels. Both runs were discarded; we rebuilt with the self-supervised vjepa2-vitl-fpc64-256 backbone and a 50% hard bound, and treat the 21 pp honest/leaky gap as a direct measurement. Several infrastructure faults — corrupted checkpoints, a snapshot-rename race, a cross-architecture ensembling bug — all traced to one root: trusting filenames instead of rebuilding each model from its stored config at inference.

4 Conclusion and Future Work

From scratch, Track A shows that data-centric fixes dominate architecture choice (+36.3 pp warm-up, +4.4 pp frame-count fix), but once fixed, a clear ranking emerges: TSM beats VFL by +3.0 pp and a $2.1\times$ deeper backbone adds +3.4 pp (42.85% val_dir). A five-model ensemble reaches 46.75% / 0.4785 Kaggle; leave-one-out shows backbone diversity beats family diversity. Track B shows that progressive unfreezing with LLRD enables effective fine-tuning of V-JEPA 2 [1] / VideoMAE [8] on a constrained budget, and that uniform diversity beats single-family or learned-weight ensembling (0.6586 Kaggle). Both tracks share the same wall: SSv2’s real-vs-

pretended and direction labels require the withheld outcome that four frames cannot supply.

Future work. (1) 10-crop TTA for TSM-ResNet50 ($\sim +2$ pp, too costly pre-deadline); (2) class rebalancing and trajectory-prediction heads for the holding/pretending sinks; (3) a second pseudo-label round with temperature-scaled thresholds (returns already flat-ten at $n=15$).

Contributions

A. Signoretti built Track A (models, training framework, ablations, ensemble). F. Guillaumey built Track B (re-extraction, pseudo-labelling, ensemble, integrity audit). Both authors wrote this report.

Figures

All figures are referenced from the main text (Secs. 2.1–3.3).

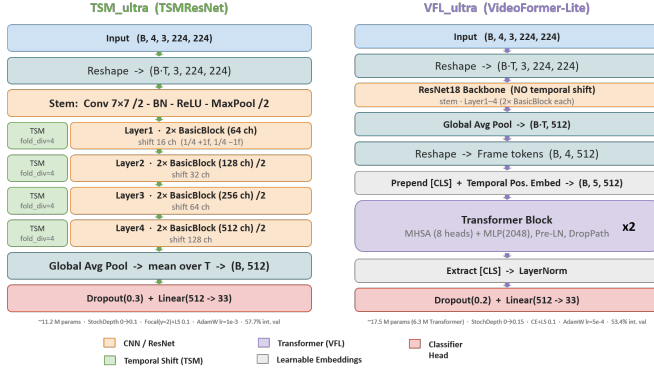


Figure 1. The two complementary Track A temporal mechanisms. TSM mixes time locally via parameter-free channel shifts in every ResNet block, before spatial pooling; VideoFormer-Lite mixes time globally via [CLS]-token attention over pooled frame descriptors. The controlled comparison in Sec. 3.1 shows the former wins by +3.0 pp at matched backbone size (Sec. 2.1).

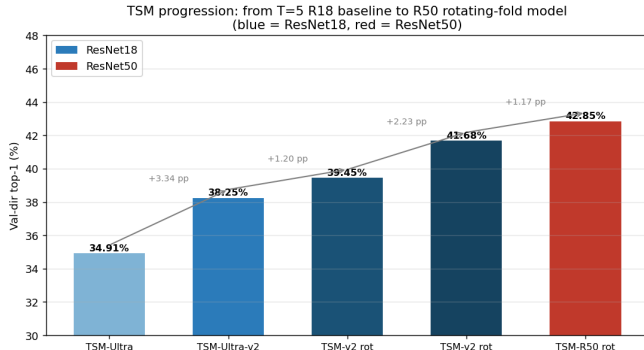


Figure 2. TSM backbone and frame-count progression on val_dir. The $T=8$ ResNet50 attempt regresses because interpolation duplicates frames; restoring $T=4$ and scaling to ResNet50 with rotating folds instead adds +3.4 pp over the matched ResNet18, reaching the new single-model best of 42.85% (Sec. 2.1).

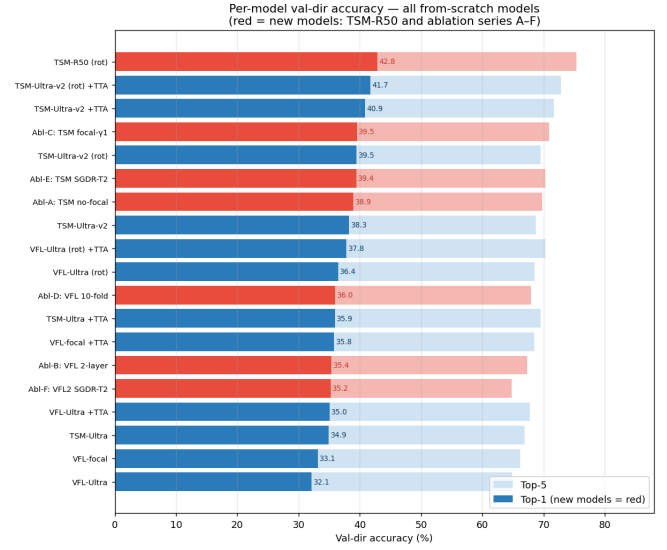


Figure 3. Per-model val_dir top-1/top-5 (Track A). Correcting the frame count (TSM-Ultra→TSM-Ultra-v2) and then scaling the backbone to ResNet50 (rotating folds) are the two largest per-model jumps; the $T=8$ ResNet50 scale-up regresses (Sec. 3).

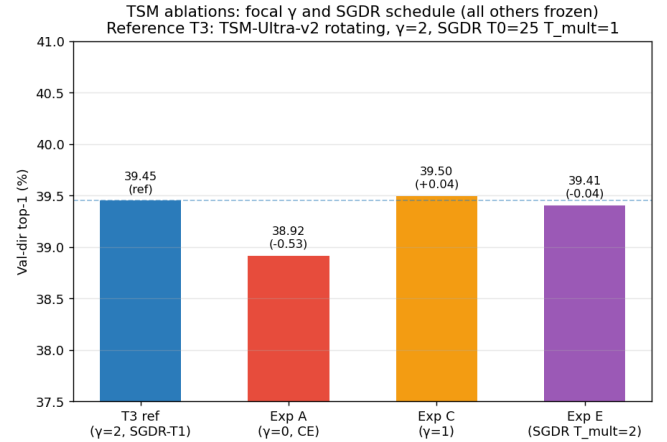


Figure 4. TSM ablations: focal- γ sweep and SGDR restart shape (all other settings frozen). Any focal weight beats CE by ≈ 0.5 pp; $\gamma=1$ and $\gamma=2$ are within noise; SGDR T_{mult} has negligible effect (< 0.05 pp) (Sec. 3.1).

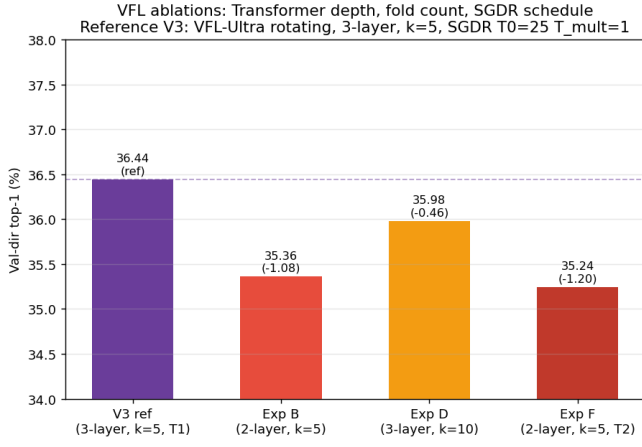


Figure 5. VFL ablations: Transformer depth (2 vs. 3 layers, -1.08 pp for 2 layers), fold count (5 vs. 10, -0.46 pp for 10 folds — VFL is data-saturated at $k=5$), and SGDR restart shape (≤ 0.12 pp effect) (Sec. 3.1).

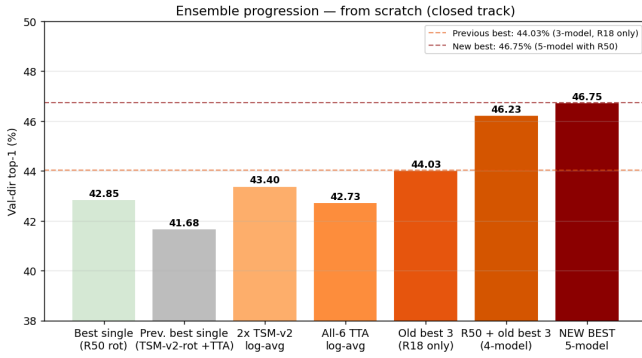


Figure 6. Track A ensemble vs. its components. Log-softmax late fusion of two TSM-ResNet18 variants, one VFL variant (all +TTA), the new TSM-ResNet50 and a CE-trained TSM ablation (no TTA) reaches 46.75% val_dir and 0.4785 public / 0.4547 private on Kaggle — our best closed-track submission (Sec. 2.1.4).

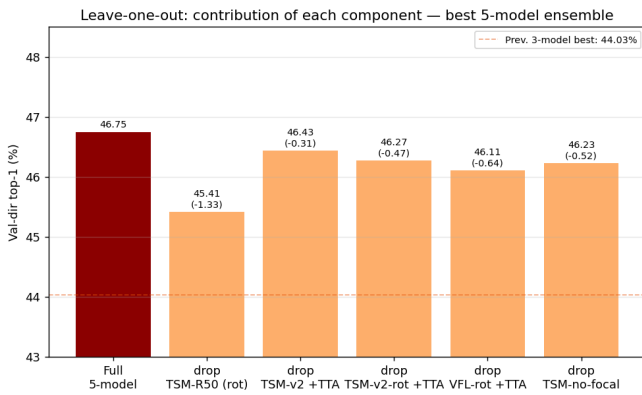


Figure 7. Track A leave-one-out study on the 5-model ensemble (val_dir top-1). The ResNet50 backbone is the dominant contributor (-1.33 pp), ahead of the architecturally distinct VFL (-0.64 pp): backbone diversity outweighs family diversity here (Table 3).

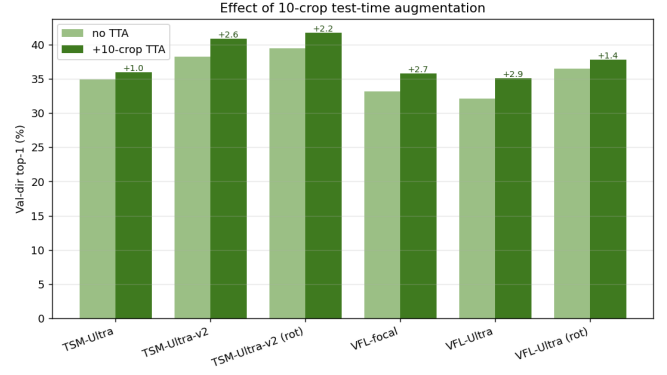


Figure 8. Effect of 10-crop spatial TTA on Track A models. TTA adds ~ 1 pp on the strongest checkpoint and is essential at the ensemble level, where independent random errors from different crops average out (Sec. 2.1).

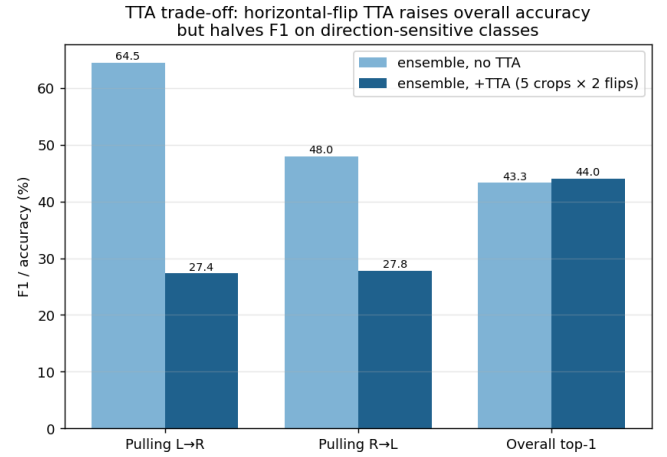


Figure 9. TTA directional-class trade-off. Horizontal-flip TTA improves most classes but hurts a few direction-encoded ones (Pulling left→right vs. right→left) even after the flip-aware logit remap, because mirror evidence is intrinsically ambiguous when motion is short.

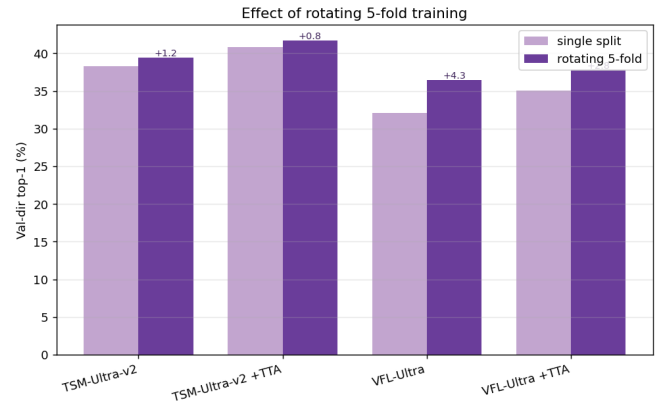


Figure 10. Rotating-fold training. Re-training the strongest checkpoints on every train/val fold yields snapshots that share weights but disagree on the boundary, contributing uncorrelated errors that the log-softmax ensemble exploits; the gain is larger for VFL ($+4.3$ pp) than for TSM ($+1.2$ pp) (Sec. 2.1.4).

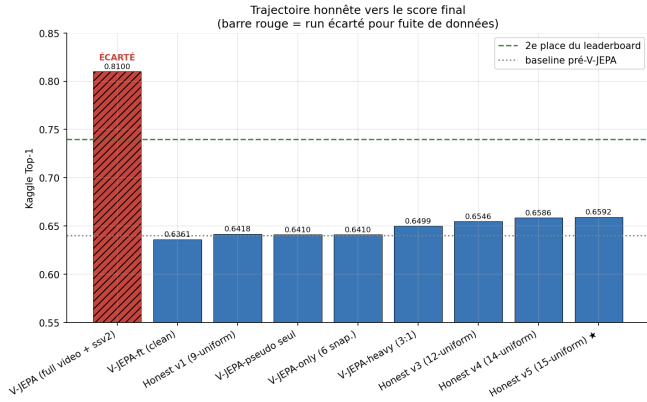


Figure 11. Track B honest Kaggle progression. The discarded leaky submission (0.81, red) is plotted alongside every clean ensemble; each gain comes from adding architecturally distinct members, not from re-weighting. The dashed line marks the second-place public reference (0.74); the gap to it bounds what self-supervised pretraining + diverse ensembling buys us (Sec. 3, Table 4).

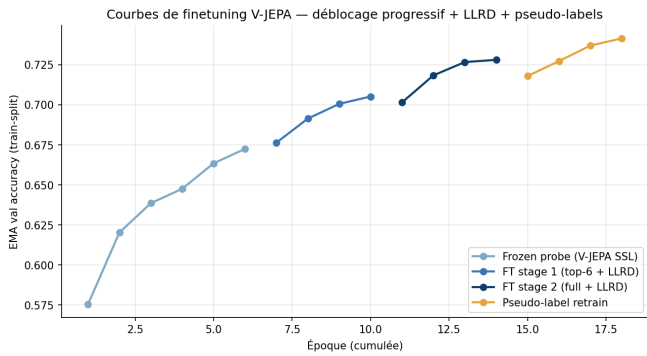


Figure 12. V-JEPA progressive unfreezing + LLRD (Track B internal val). Each phase warm-starts above the previous plateau: frozen probe (0.55 $\rightarrow$ 0.67), then top-6 unfrozen + LLRD (0.68 $\rightarrow$ 0.71), then full encoder (0.70 $\rightarrow$ 0.73), then pseudo-label retrain (0.72 $\rightarrow$ 0.74). The staircase pattern is the visual signature that LLRD has preserved each phase’s feature geometry (Sec. 2.2).

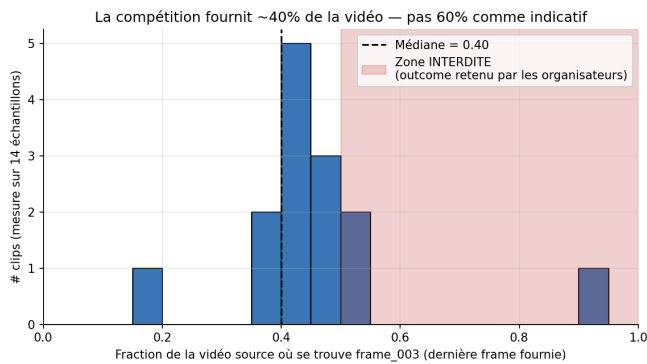


Figure 13. Where shipped frame_003 sits in the source SSv2 clip (fraction). The dashed line at 0.40 is the median end-of-window; the shaded region [0.50, 1.00] is the withheld outcome — our re-extraction is hard-bounded to the un-shaded region (Sec. 3.4).

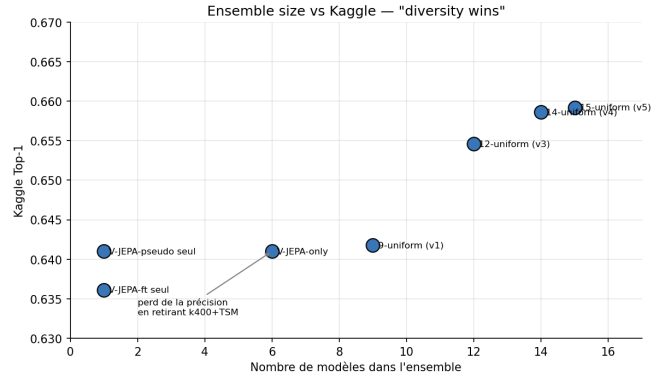


Figure 14. Track B Kaggle vs. ensemble size. Returns from adding diverse honest members are still positive at $n=14$; at $n=6$, dropping K400 + TSM hurts despite their lower individual scores, because their uncorrelated errors disappear from the average (Table 5).

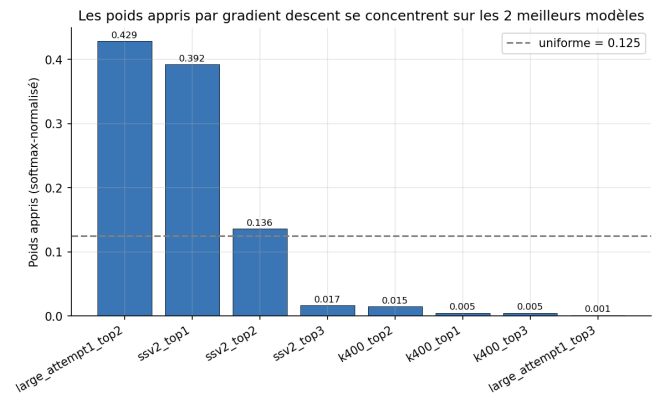


Figure 15. Learned ensemble weights collapse onto the two strongest members ($\sim 82\%$ of mass on large_attempt1_top2 and ssv2_top1), effectively zeroing K400 contributions. This is exactly why a learned weighting fails to beat uniform: it discards the very decorrelation signal that diverse weaker members provide (Sec. 3).

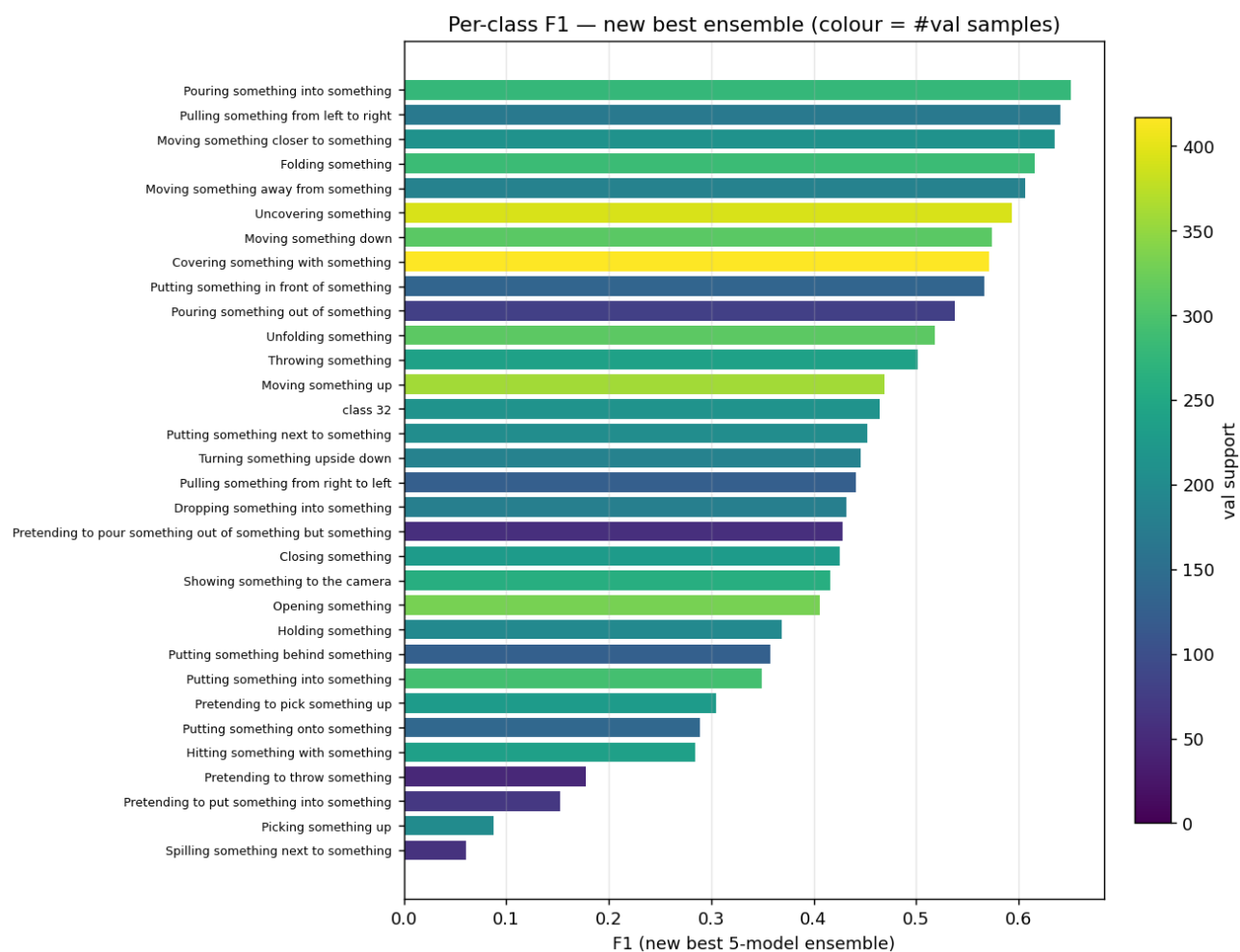


Figure 16. Per-class F1 (5-model ensemble, val_dir). Directional single-object motions score highest; “pretending” and direction-ambiguous classes score lowest (Sec. 3.3).

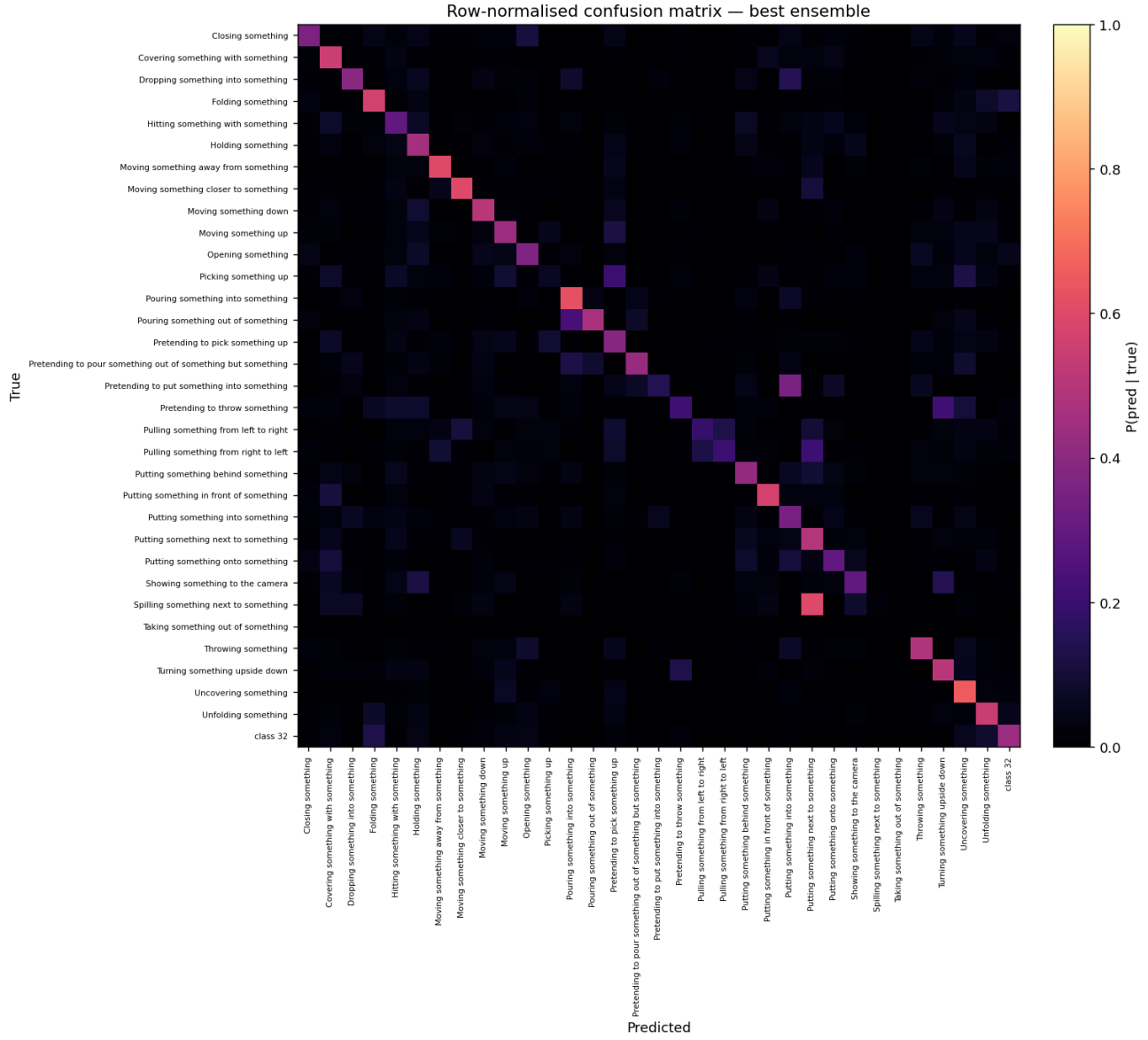


Figure 17. 33×33 confusion matrix (5-model ensemble, val_dir). Off-diagonal mass concentrates on semantically adjacent pairs (real vs. pretended; fold vs. unfold; motion vs. holding).

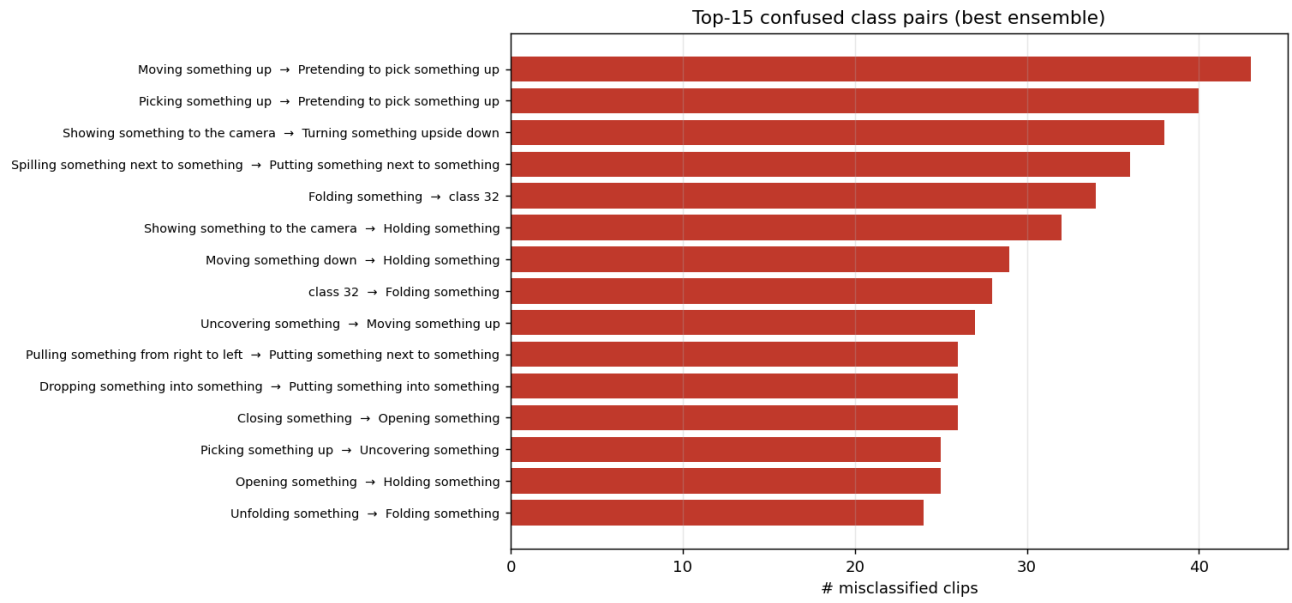


Figure 18. Most-confused class pairs. The dominant errors are intent (real vs. pretended) and temporal-direction confusions (Sec. 3.3).

References

- [1] Adrien Bardes, Quentin Garrido, Jean Ponce, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. arXiv preprint arXiv:2506.09985, 2025. [1](#), [3](#), [5](#)
- [2] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is Space-Time Attention All You Need for Video Understanding? In ICML, 2021. [1](#), [2](#)
- [3] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In ICLR, 2021. [2](#)
- [4] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The “Something Something” Video Database for Learning and Evaluating Visual Common Sense. In ICCV, 2017. [1](#)
- [5] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. In CVPR, 2016. [2](#), [3](#)
- [6] Ji Lin, Chuang Gan, and Song Han. TSM: Temporal Shift Module for Efficient Video Understanding. In ICCV, 2019. [1](#), [2](#), [3](#)
- [7] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal Loss for Dense Object Detection. In ICCV, 2017. [2](#)
- [8] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training. In NeurIPS, 2022. [1](#), [3](#), [5](#)
- [9] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On Layer Normalization in the Transformer Architecture. In ICML, 2020. [2](#)
- [10] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. CutMix: Regularization Strategy to Train Strong Classifiers with Localizable Features. In ICCV, 2019. [2](#)
- [11] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond Empirical Risk Minimization. In ICLR, 2018. [2](#)